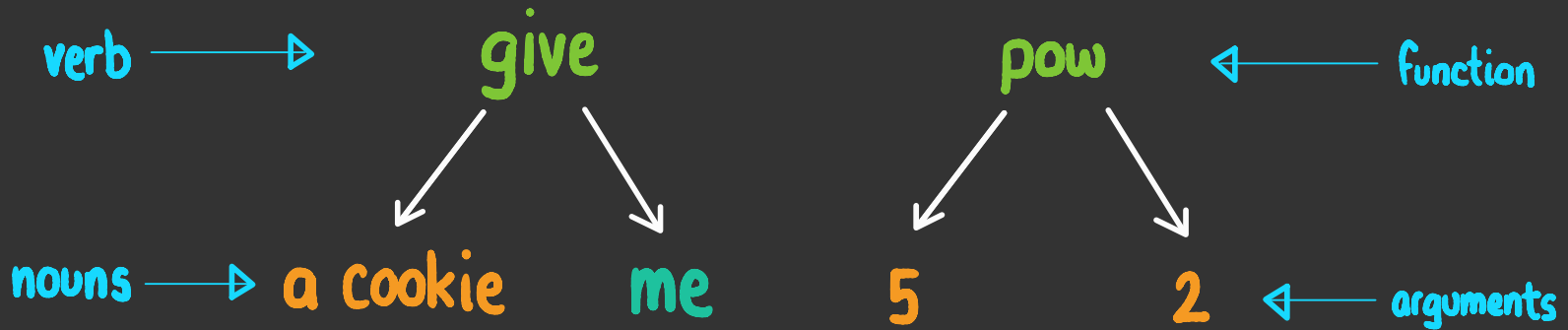


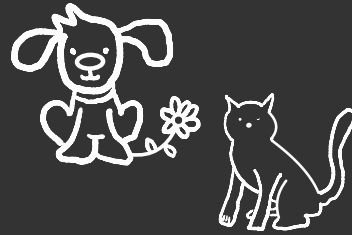
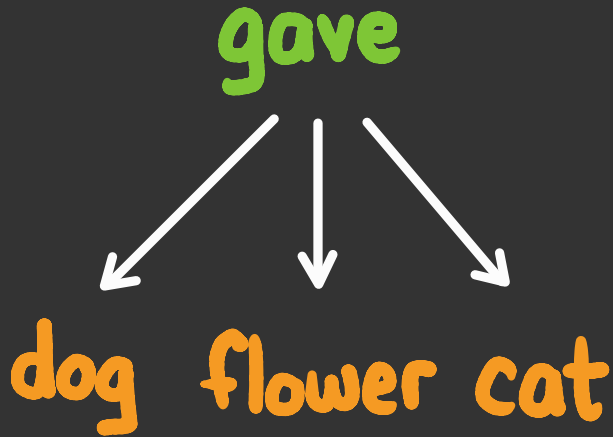
# expression trees

CSCI  
134

# these are examples of expression trees

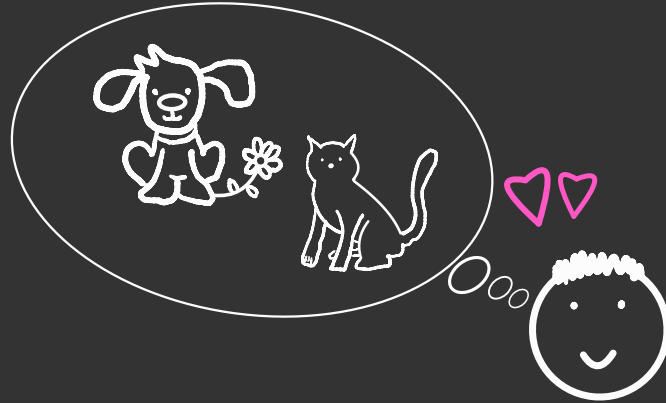
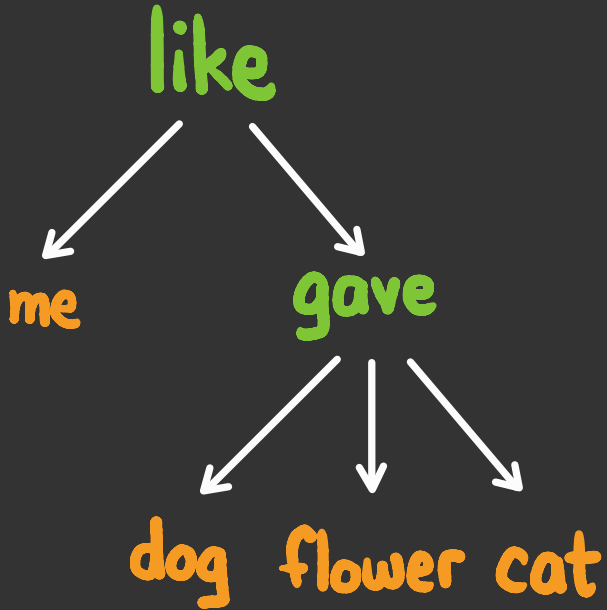


but expression trees are often more **complex**



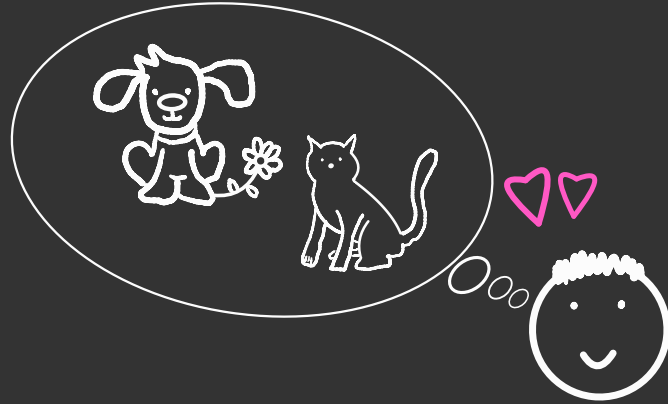
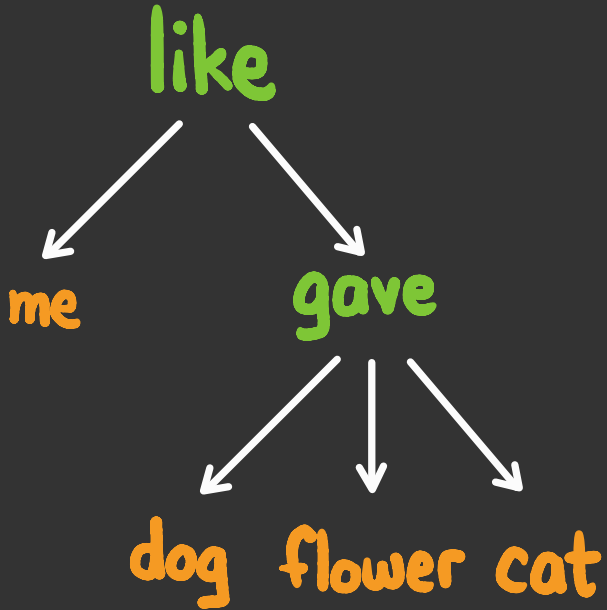
"the dog gave a  
flower to the cat"

e.g. expression trees can be used as arguments



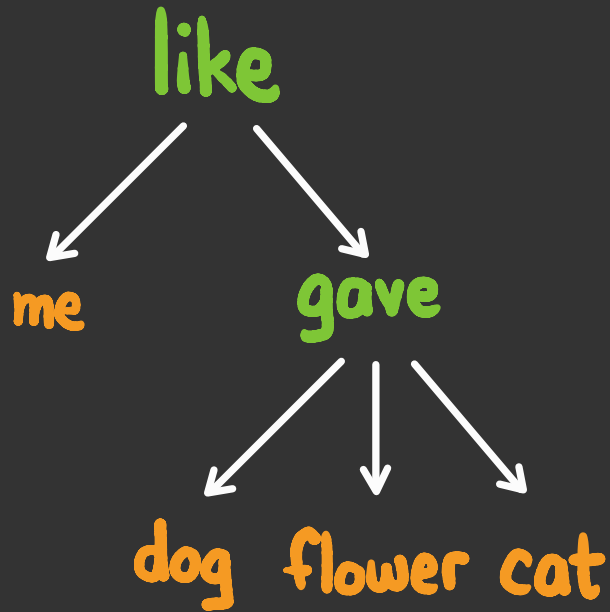
how might this be expressed  
as an english sentence?

e.g. expression trees can be used as arguments

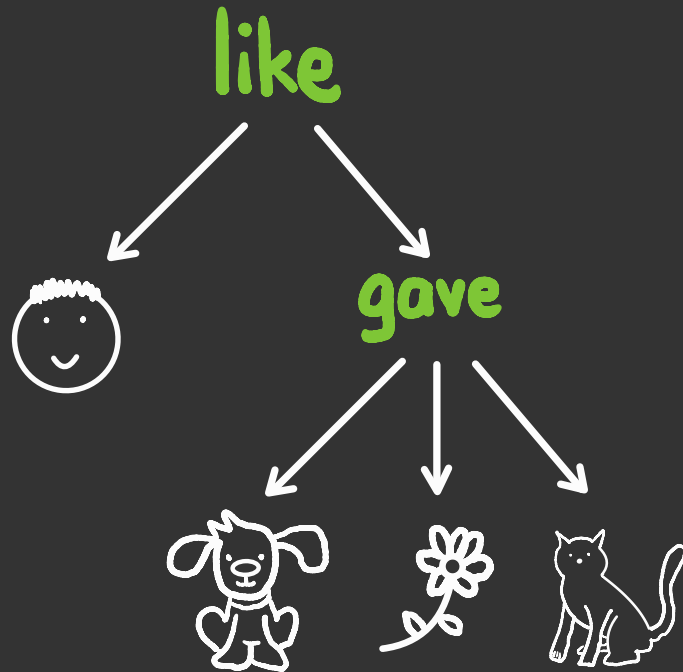


"I like that the dog gave  
a flower to the cat"

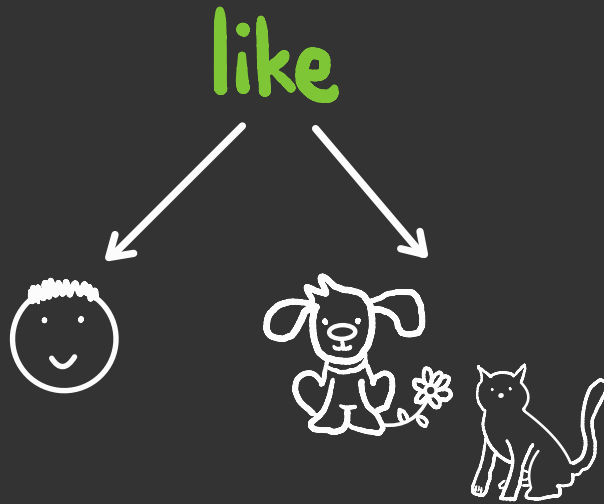
the meaning is constructed "bottom-up"



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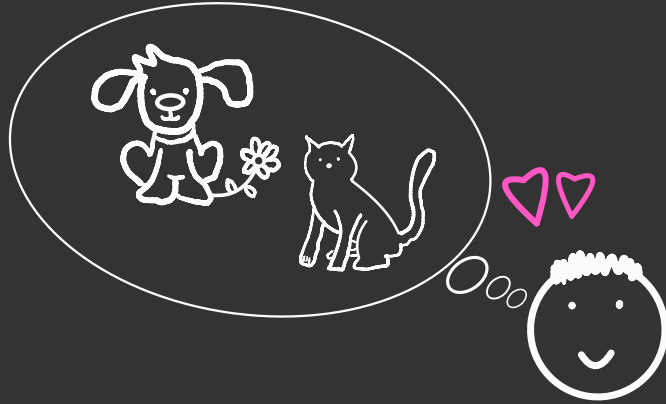


the meaning is constructed "bottom-up"

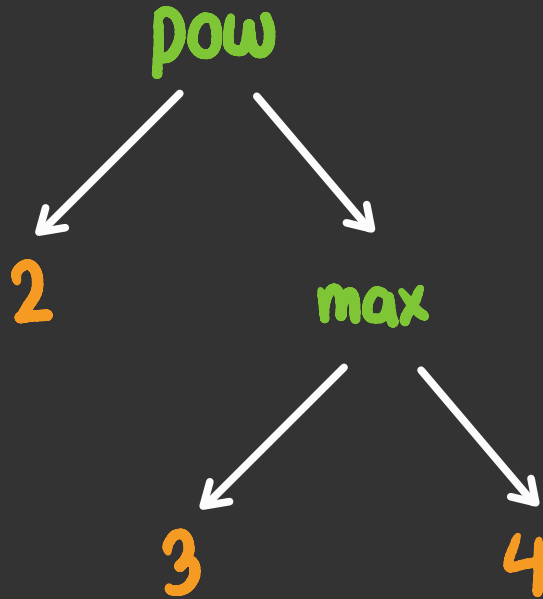




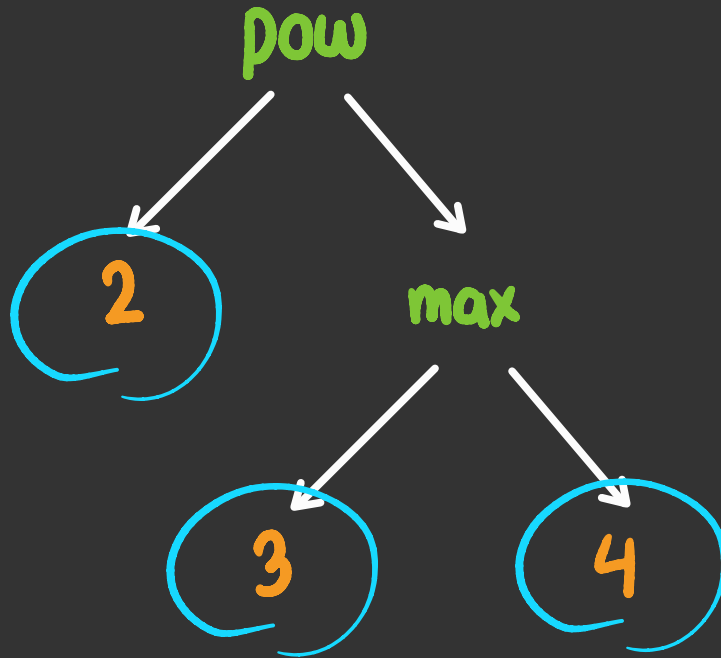
this is called **evaluating** an expression tree



let's evaluate a **python** expression tree

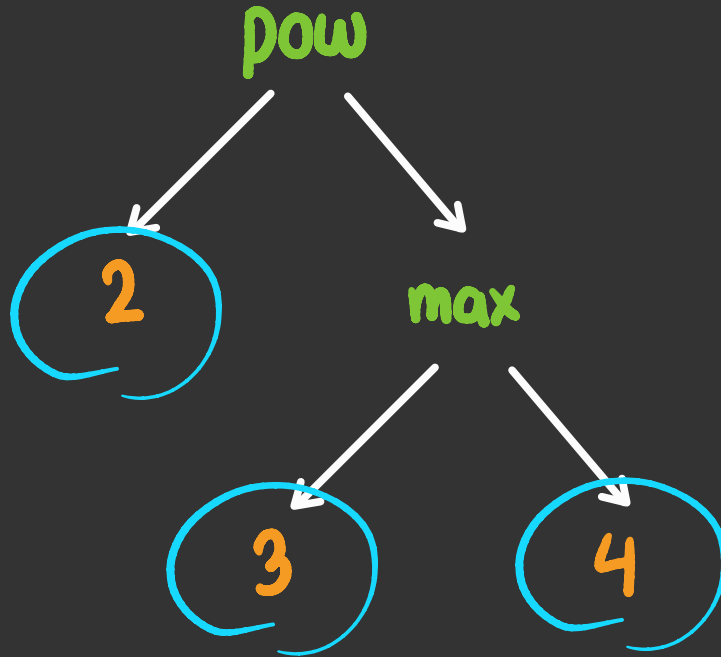


at the bottom of the tree are **primitive expressions**

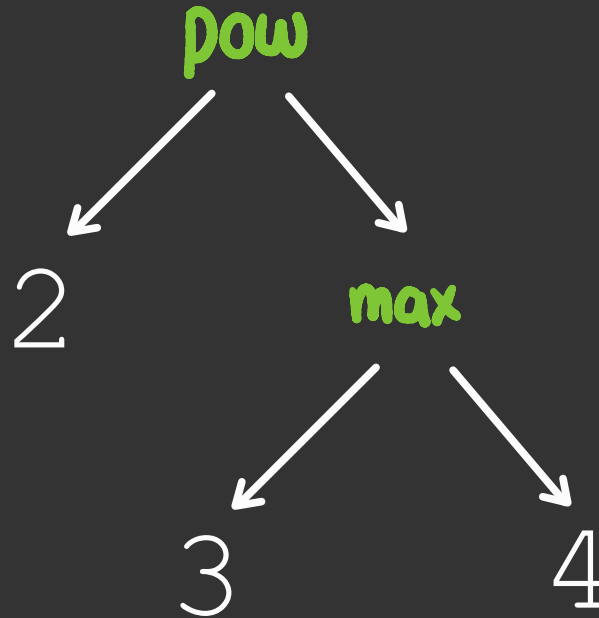


in this case,  
**ints**

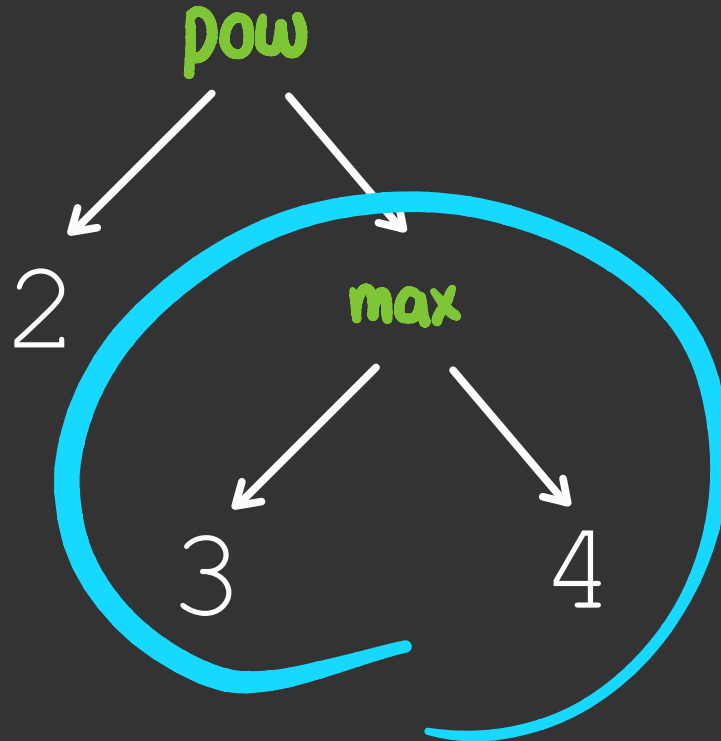
the value of a **primitive expression** is just itself



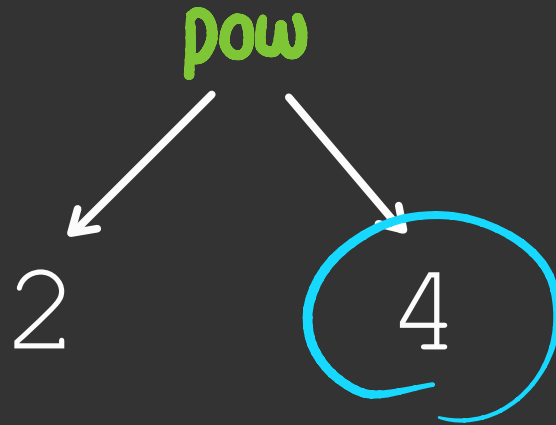
the value of a **primitive expression** is just itself



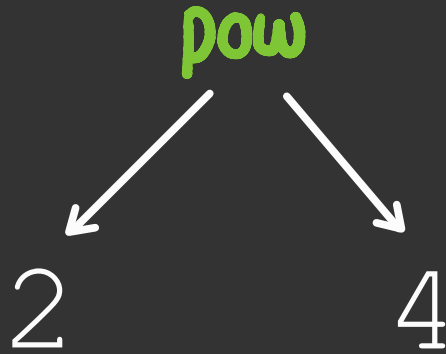
next we have a function with two arguments



the maximum of 3 and 4 is 4



now we have another function with two arguments

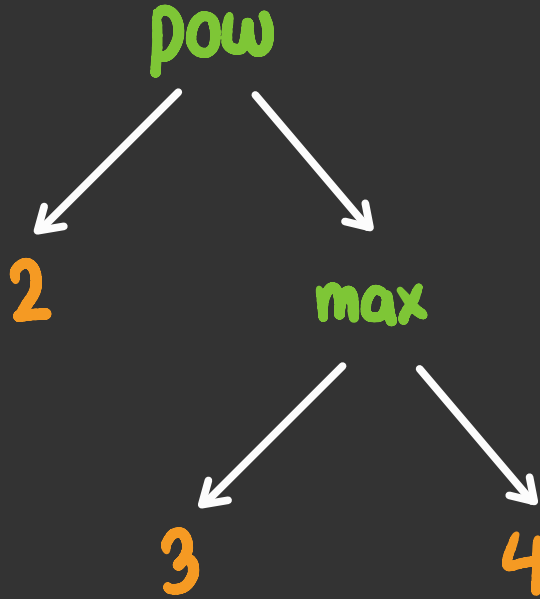




2 to the power of 4 is 16

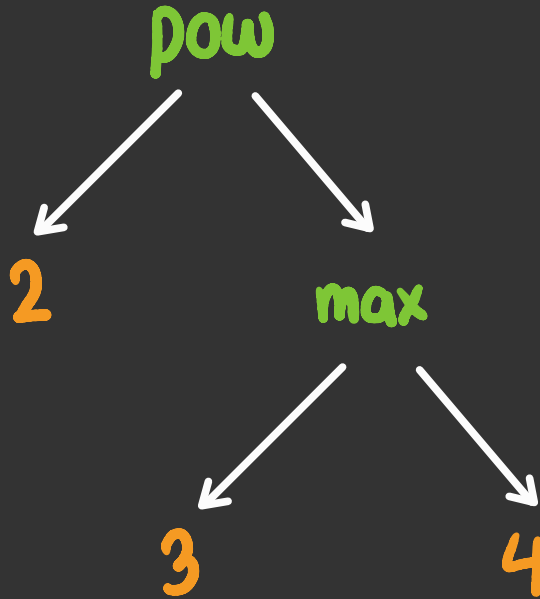
16

therefore, the value of the expression tree is 16



16

the type of an expression tree is  
the type of its value



16

so the type of this  
expression tree is  
int